Morphology-Aware Complexity and LLM-Based Simplification for Tamil

Proposed extension of the Singh et al. (2016) replication — Phase 3

Abstract

Computational readability and text-simplification research has largely been built around English and other morphologically simple languages, where a word's orthographic length is a reasonable proxy for how much information it carries. Tamil, a morphologically rich, agglutinative Dravidian language, breaks this assumption: a single Tamil word can carry case, tense, negation, and agreement information that English spreads across an entire clause, so word-length-based complexity metrics — including the one used by Singh et al. (2016), whose method this thesis already replicated and extended with GPT-2 surprisal for English in Phase 2 — are structurally mismatched to Tamil. This thesis proposes and tests a **morphology-aware complexity metric** for Tamil, combining morpheme-level segmentation (via the open-source `ThamizhiMorph` analyser) with language-model surprisal, and compares it against a naive word-level metric ported directly from the English approach. The metric is then used to drive and evaluate an **LLM-based Tamil sentence simplification** system: a capable language model generates simplified Tamil rewrites of complex sentences, and simplification quality is scored using both metrics, with results validated against a small native-speaker judgment study. The central claim is that the morphology-aware metric tracks perceived simplification quality more closely than the naive word-level metric — demonstrating that complexity metrics designed for English do not transfer safely to agglutinative languages, and providing Tamil with both a validated complexity metric and a working simplification pipeline, neither of which currently exists.

1. Motivation

- Singh et al. (2016)'s word-length feature is already an imperfect proxy in English — this thesis's own investigation found the paper's Table 1 ($r=0.765$) and Table 2 ($R^2=0.267$) for word length alone are internally inconsistent (see `phase1_replication/README.md`, Improvement Roadmap #7). If word length is already unstable as a complexity signal in English, it is reasonable to expect it fails more seriously in a morphologically rich language like Tamil, where word length conflates orthographic length with stacked grammatical morphemes.
- This reframes the open question from this thesis's Phase 1/2 work (why does word length behave strangely?) into a testable hypothesis for Phase 3: word length is not just noisy, it is the *wrong unit of analysis* for agglutinative languages — morphemes are.
- A working, open-source Tamil morphological analyser exists (`ThamizhiMorph`, FST-based, GitHub: `sarves/thamizhi-morph`), making morpheme-level features genuinely buildable, not just theoretical.
- LLM-based text simplification is an active, modern NLP direction (cf. the TSAR workshop series) — grounding it in a Tamil-specific, linguistically motivated metric (rather than porting an English-style metric unchanged) is what makes this a contribution rather than a port.

2. Related Work / Novelty Check

Searched specifically for prior Tamil work before committing to this direction:

- **No Tamil eye-tracking-while-reading corpus** — the multilingual MultiplEYE initiative does not yet cover Tamil, so ground-truth reading-time data (System 1 in the original paper) is unavailable.
- **No Tamil readability/complexity classifier or simplification system found** in DravidianLangTech (the main Tamil/Dravidian NLP workshop, ACL Anthology) — active Tamil work there covers sentiment, hate speech, ASR, and code-mixing, but not readability or complexity.
- **Closest Indic-language precedent**: a 2023 ACM TALLIP paper building a machine-learning readability model for **Gujarati** — confirms this general style of work is recognized and publishable for an Indic language, while leaving Tamil specifically unaddressed.
- One paper appeared promising but is **not** real prior art on closer inspection: Gooding et al. (2021, Google Research), "Predicting Text Readability from Scrolling Interactions," includes Tamil-first-language readers — but only as a demographic variable while those readers read **English** text; no Tamil text or Tamil-specific model was built.
- No published Tamil AoA (age-of-acquisition) norms and no mature Tamil constituency parser/treebank at scale were found — this is precisely why the original paper's AoA-based and parse-tree-based features (embedding depth/difference, hierarchical structure, entropy reduction) are not used in this proposal; morphological features are the substitute this thesis leans into instead.
- Caveat: this is a search-tool-based check, not exhaustive (paywalled papers and non-indexed regional proceedings may exist) — worth a direct Google Scholar/ICON-proceedings check before finalizing the novelty claim in the actual thesis writeup.

3. Research Questions

1. Does a morpheme-level complexity metric (morpheme count + morpheme- level surprisal) predict perceived Tamil sentence difficulty better than a word-level metric (word length + word-level GPT-2 surprisal, ported directly from the English approach)?
2. Can an LLM-generated Tamil simplification be reliably evaluated using the morphology-aware metric, and does it agree better with human judgments than the naive word-level metric?

4. What's Reused vs. What's New

Component	Status
Word-level GPT-2 surprisal pipeline	Reused directly from Phase 2 (`compute_gpt2_surprisal.py`), swapped to a Tamil GPT-2 model

Component	Status
Word length / sentence length features	Reused as the "naive" baseline metric to beat
AoA, embedding depth/difference, hierarchical structure (h1–h8), entropy reduction	**Dropped** — no Tamil AoA norms or constituency parser exist; explicitly out of scope, not silently ignored
Morpheme segmentation + morpheme count	**New** — via `ThamizhiMorph`
Morpheme-level surprisal	**New** — feed morpheme sequences (not word sequences) through a Tamil LM
LLM-based simplification generator	**New** — prompted, not trained from scratch
Human judgment validation study	**New** — small native-speaker panel

5. Plan (phased)

Phase A — Morphology-aware metric (2 weeks) 1. Integrate `ThamizhiMorph` to segment Tamil sentences into morphemes. 2. Build morpheme-level surprisal (morpheme sequence through a Tamil LM) alongside the existing word-level GPT-2 surprisal pipeline. 3. Define the composite morphology-aware complexity score.

Phase B — Simplification generator (1–2 weeks) 4. Prompt-engineer an LLM to simplify complex Tamil sentences (start with a small hand-picked set of complex Tamil sentences — news editorials are a convenient source). 5. Generate simplified candidates; filter for basic fluency/validity.

Phase C — Evaluation (1–2 weeks) 6. Score original vs. simplified pairs with both metrics (naive word-level vs. morphology-aware). 7. Run a small human judgment study (~10–20 native Tamil speakers, a few dozen sentence pairs) rating perceived simplification quality/ease. 8. Compare which metric's scores correlate better with human judgments — this comparison is the thesis's central empirical result.

Phase D — Writeup (1 week) 9. Report correlation results, discuss where/why the morphology-aware metric wins or loses, and package the pipeline for reuse.

Total estimated time: ~6–8 weeks — comparable in scope to the earlier System-2-port plan, but the deliverable is a genuinely new metric and system rather than a straight adaptation.

6. Tools & Resources

Need	Source
Tamil morphological segmentation	`ThamizhiMorph` (github.com/sarves/thamizhi-morph)
Tamil surprisal (word- and morpheme-level)	`abinayam/gpt-2-tamil` or `Lagstill/GPT-2-Tamil` (Hugging Face)
Simplification generation	A capable general-purpose LLM, prompted (no fine-tuning needed for MVP)
Complex source sentences	Tamil newspaper editorials / Tamil Wikipedia
Human validation	Small panel of native Tamil speakers (~10–20)

7. Risks / Open Questions

- `ThamizhiMorph`'s coverage/accuracy on newspaper-register text is unverified — needs a quick pilot check before committing Phase A's timeline.
- LLM-generated Tamil simplifications may have fluency or meaning- preservation issues that need manual filtering — budget time for this.
- Human validation panel size (~10–20) is small; frame results as an initial/pilot-scale finding, not a definitive claim — honest framing is stronger than overclaiming on a small n.
- The morphology-aware metric could turn out *not* to beat the naive one — that would still be a valid, reportable (if less exciting) thesis result, since the research question is genuinely open.

8. Expected Contribution

1. First morphology-aware text-complexity metric for Tamil, directly motivated by a weakness (word length as a proxy) this thesis already identified empirically in the English replication.
2. First Tamil sentence-simplification pipeline evaluated against a linguistically-motivated (not just ported) complexity metric.
3. A concrete empirical answer to whether English-shaped complexity metrics transfer to agglutinative languages — a comparative finding of interest beyond Tamil specifically (any agglutinative language: Turkish, Finnish, Korean, etc. faces the same mismatch).